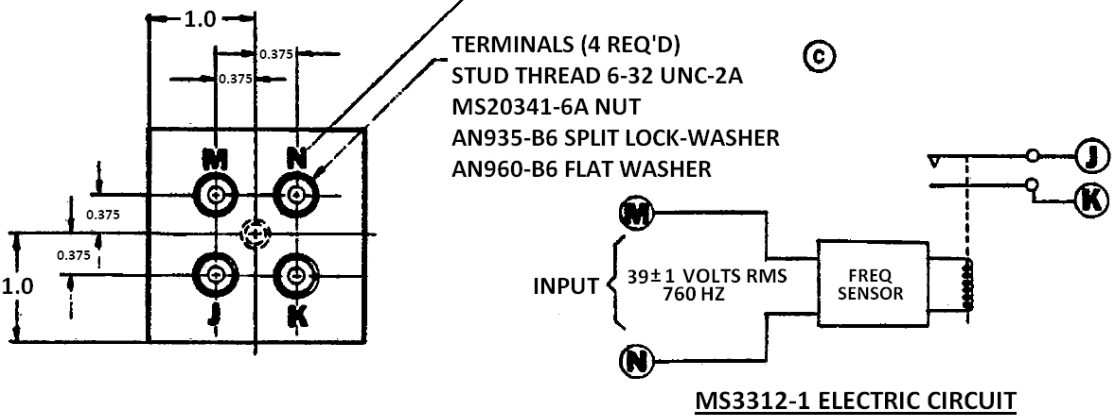
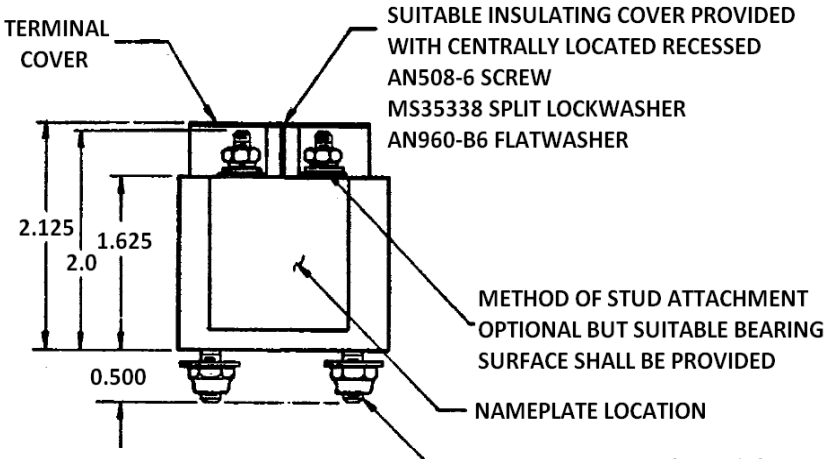
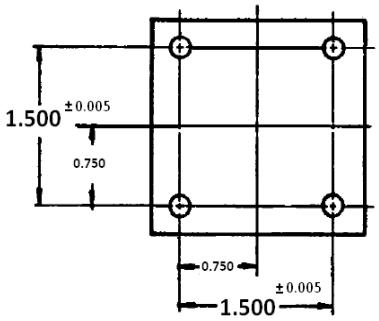
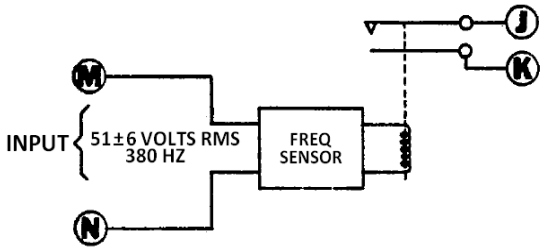


MS3312-1 ELECTRIC CIRCUIT





MAXIMUM WEIGHT: 0.5 POUND



MS3312-2 ELECTRIC CIRCUIT

MS PART NO.	INPUT POTENTIAL VOLTS	FREQUENCY RANGE HERTZ
MS3312-1	39 ± 1	725 – 775
MS3312-2	51 ± 6	362.5 – 387.5

© DENOTES CHANGES

APPROVED 9 FEB 1968 REVISED (A) 23 JUNE 1977 (B) 22 JULY 1977 (C) 30 SEPT 80

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P.A. NAVY – AS Other Cust.	TITLE **PROTECTOR, UNDERFREQUENCY F-4 AIRCRAFT**	**MILITARY STANDARD** **MS3312(AS)**
PROCUEMENT SPECIFICATION MIL-G-21480	SUPERSEDES	SHEET 1 OF 3

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NOTES:
1. DIMENSIONS IN INCHES; TOLERANCES ± 0.03125 UNLESS OTHERWISE SPECIFIED.
2. ALL OUTLINE DIMENSIONS ARE LIMITING DIMENSIONS ONLY.

REQUIREMENTS:

1. ANY PROTRUSION SUCH AS SCREW AND RIVET HEADS AND BOSSES SHALL BE CONTAINED WITHIN THE OUTLINE SHOWN.
2. THE UNDERFREQUENCY PROTECTOR SHALL BE FINISHED IN A COLOR CONFORMING TO MIL-STD-595 COLOR NUMBER 17875. THE FINISH SHALL BE UNAFFECTED BY THE ENVIRONMENTAL CONDITIONS SPECIFIED HEREIN OR FLUIDS NORMALLY USED IN SERVICING AIRCRAFT.
3. THE UNDERFREQUENCY PROTECTOR SHALL BE COMPATIBLE WITH THE AC GENERATING SYSTEM DESCRIBED IN MS90302. RELAYS INCORPORATED IN THE UNDERFREQUENCY PROTECTOR SHALL COMPLY WITH MIL-R-5757.
4. a. THE MS3312-1 UNDERFREQUENCY PROTECTOR SHALL BE CAPABLE OF BREAKING THE CIRCUIT J-K (SEE WIRING DIAGRAM ABOVE) BEFORE THE INPUT FREQUENCY IS LESS THAN 725 HZ. AND SHALL CLOSE CIRCUIT J-K BEFORE THE INPUT FREQUENCY IS ABOVE 775 HZ. THERE SHALL BE A MINIMUM DIFFERENCE BETWEEN CLOSING AND OPENING OF 20HZ. ©

   b. THE MS3312-2 UNDERFREQUENCY PROTECTOR SHALL BE CAPABLE OF BREAKING THE CIRCUIT J-K (SEE WIRING DIAGRAM ABOVE) BEFORE THE INPUT FREQUENCY IS LESS THAN 362.5 HZ. AND SHALL CLOSE CIRCUIT J-K BEFORE THE INPUT FREQUENCY IS ABOVE 387.5 HZ. THERE SHALL BE A MINIMUM DIFFERENCE BETWEEN CLOSING AND OPENING OF 10 HZ.
5. THE COMPLETE REQUIREMENTS FOR PROCURING THE UNDERFREQUENCY PROTECTOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND MIL-G-21480, AS AMENDED OR DELETED BELOW. IN ALL INSTANCES SUBSTITUTE "UNDERFREQUENCY PROTECTOR" FOR "GENERATOR", "REGULATOR", "SUPERVISORY PANEL", AND "GENERATOR CONTROL UNIT".

1.1 SCOPE. DELETE AND ADD: THIS SPECIFICATION COVERS THE REQUIREMENTS FOR AN UNDERFREQUENCY PROTECTION UNIT TO BE USED WITH AN AIRCRAFT CONSTANT FREQUENCY (400 HZ) ELECTRIC POWER SYSTEM.

3.4.10.1 VOLTAGE SETTING: DELETE AND ADD:

(a) THE MS3312-1 UNDERFREQUENCY PROTECTOR SHALL HAVE 39 ± 1 VOLT APPLIED BETWEEN PINS M AND N.

(b) THE MS3312-2 UNDERFREQUENCY PROTECTOR SHALL HAVE 51 ± 6 VOLT APPLIED BETWEEN PINS M AND N.

3.4.14 IDENTIFICATION OF PRODUCT: ADD: NAMEPLATE SHALL BE IN ACCORDANCE WITH MIL-N-18307.

3.5.7.1 FALSE TRIPS: CHANGE TO READ: "NO TRIPPING OF THE GENERATOR SYSTEM SHALL BE CAUSED BY THE BUILD-UP, LOAD SWITCHING, INPUT SPEED ACCELERATION OR DECELERATION, DURING EXTERNAL POWER APPLICATION, OR FAULT CLEARING OUTSIDE THE OPERATING BAND OF THE UNDERFREQUENCY PROTECTOR."

3.5.7.4 UNDERFREQUENCY: CHANGE TO READ: "THE UNDERFREQUENCY PROTECTOR SHALL PROVIDE MEANS OF PREVENTING A FREQUENCY OF LESS THAN 362.5 HZ BEING IMPRESSED ON THE LOAD BUS FOR A PERIOD IN EXCESS OF 0.1 SECOND. THE DEVICE SHALL HAVE A MAXIMUM PICKUP OF 387.5 HZ."

4.2.1.1 ADD: QUALIFICATION TESTS SHALL BE SUPPLEMENTED WITH A MINIMUM 100-HOUR FAILURE-FREE FLIGHT TEST BEFORE CONSIDERATION IS GIVEN TO INCORPORATING THE ITEM ON THE QUALIFIED PRODUCTS LIST.

© 4.5.14 DELETE (a), (b), (c), (d), AND (e) AND ADD:

(a) WITH THE UNDERFREQUENCY PROTECTOR CONNECTED INTO THE AC GENERATING SYSTEM (MS90302), IT SHALL BE CAPABLE OF OPERATION WITHIN 0.5% OF REQUIREMENT 4 WHEN SUBJECTED TO THE FOLLOWING AMBIENT TEMPERATURES AND ALTITUDES: SEA LEVEL +125˚C AND 50,000 FEET AT -55˚C.

(b) THE UNDERFREQUENCY PROTECTOR SHALL BE SUBJECTED TO VIBRATION TESTS, PARAGRAPH 4.9 OF MIL-E-81910(AS), AFTER WHICH IT SHALL MEET THE REQUIREMENTS OF THE REQUIREMENT 4 ABOVE.

6.1 DELETE AND ADD: INTENDED USE: THE UNDERFREQUENCY PROTECTOR MAY BE INSTALLED IN THE SYSTEM (MS90302) SO AS TO AUTOMATICALLY DE-ENERGIZE THE GENERATOR DURING SHUT-DOWN AND/OR CONSTANT SPEED DRIVE MALFUNCTION SO THAT LOW-FREQUENCY POWER WILL NOT BE APPLIED TO THE AIRCRAFT BUS.

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DELETE ALL REFERENCE AND REQUIREMENTS OF:

3.4.6	DELETE
3.4.7	DELETE AND ALL SUBPARAGRAPHS
3.4.8	DELETE AND ALL SUBPARAGRAPHS
3.4.9	DELETE AND ALL SUBPARAGRAPHS
3.4.10	DELETE
3.5	DELETE
3.5.1	DELETE AND ALL SUBPARAGRAPHS
3.5.2	DELETE AND ALL SUBPARAGRAPHS
3.5.3	DELETE
3.5.4	DELETE
3.5.5	DELETE
3.5.6	DELETE
3.5.7.2	DELETE
3.5.7.3	DELETE
3.5.7.6	DELETE
3.5.7.7	DELETE
4.3	DELETE AND ALL SUBPARAGRAPHS
4.4	DELETE AND ALL SUBPARAGRAPHS
4.5.2	DELETE
4.5.3	DELETE
4.5.4	DELETE AND ALL SUBPARAGRAPHS
4.5.6	DELETE
4.5.8	DELETE AND ALL SUBPARAGRAPHS
4.5.9	DELETE
4.5.10	DELETE
4.5.11	DELETE
4.5.13	DELETE
4.5.16	DELETE
4.5.19	DELETE
4.5.23.1.1	DELETE
4.5.26	DELETE
4.5.27	DELETE
4.5.28	DELETE
4.5.29	DELETE

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

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